Al-Balqa Applied University

جامعة البلقاء التطبيقية

Part 1: Multiple choice questions

Q1: The linear system

$$x + y - z = 2$$
$$x + 2y + z = 3$$
$$x + y + (a^2 - 5)z = a$$

has unique solution if $a \in$

A) $\mathbb{R} - \{-2, 2\}$ B) $\{-2\}$ C) $\{2\}$ D) $\{-2, 2\}$

Q2: Let $u = (2, 0, 0)$. Then the value(s) of the scalar k such that $\|ku\| = 20$ is

A) $\mathbb{R} - \{-6, 6\}$ B) $\{-4\}$ C) $\{5\}$ D) $\{-5, 5\}$

Q3: Let $det(A) = 2$ and $det(B) = -2$, A and B are 2×2 matrices. Then the value of $det(2A^tB^{-1})$ is :

A) 4 B) -4 C) 16 D) -16

, Q4: If $b \neq 0$ $A = \begin{bmatrix} b & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & b \end{bmatrix}$ $B = \begin{bmatrix} -3 & 2 & 1 \\ 0 & b & 1 \\ 0 & 0 & b \end{bmatrix}$, then $det(2AB^{-1}) =$

A) 8 B) -6 C) 1 D) -1

Q5: The value of k that makes $\begin{pmatrix} k & 2k \\ 1 & k \end{pmatrix}$ singular is:

A) $\{0, 3\}$ B) $\{0, 2\}$ C) 3 D) 2

, Q6: If $A = \begin{bmatrix} c & 2 & 0 \\ 0 & c & 1 \\ 0 & 0 & c \end{bmatrix}$, and $det(2A) = 64$, then the value of $2c =$

A) 2 B) 4 C) 6 D) 8

, Q7: If $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 1 \end{bmatrix}$, then $M_{22} =$

A) 1 B) -1 C) 3 D) -3

Q8: If $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 2 \\ 2 & 1 & 1 \end{bmatrix}$, then $C_{22} =$

A) 1 B) -1 C) 3 D) -3

Part two: True or False questions:

Q1: The transpose of an upper triangular matrix is an upper triangular. (TRUE\ FALSE)

Q2: $det(A+B) = det(A) + det(B)$. (TRUE\ FALSE)

Q3: If $det(2A) = 0$ **, for a square matrix of order** n**, then** A **is singular (not invertible) matrix.** (TRUE\ FALSE)

Q4: If the dot product of two vectors equals zero, then one of the vectors is the zero vector.
(TRUE\ FALSE)

$2A = 0$

$2^n \det$

$2 \det A$

Part three: Written questions. (6 marks)

Solve the following linear system of equations using Cramer's rule.

$$x + y = Z + 0$$
$$y + z = -3 \quad y = -3 - z$$
$$x = z + 1$$
$$x - z = 1$$

Solution:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & -3 \\ 1 & 0 & -1 & 1 \end{array}\right]$$

$AX = B$

$X = A^{-1}B$

$$B = \begin{bmatrix} 0 \\ -3 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$X_1 = \frac{\det A_1}{\det A} = \frac{\begin{bmatrix} 0 & 1 & -1 \\ -3 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}}{1} = \frac{-1}{1}$$

$$X_2 = \frac{\det A_2}{\det A} = \frac{\begin{bmatrix} 1 & 0 & -1 \\ 0 & -3 & 1 \\ 1 & 1 & -1 \end{bmatrix}}{1} = \frac{-1}{1}$$

$$X_3 = \frac{\det A_3}{\det A} = \frac{\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -3 \\ 1 & 0 & 1 \end{bmatrix}}{1} = \frac{-2}{1}$$

$-1 + 1$

$\det = (1)(2) - (0)(1) + (-1)(1)$

$\det A = 1$

$X_1 = -1 \quad X_2 = -1$

$X_3 = -2$